LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory
Wednesday January 13, 2016
Duration: 8.00–13.00

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

1. Calculate the Legendre symbols

$$(93/191) \text{ and } (93/223).$$

(Here 191 and 223 are primes.)

2. Let p be a prime and let $1 < k < p-1$. Show that

$$(p-k)!(k-1)! \equiv (-1)^k \pmod{p}.$$

3. Prove that for every positive integer n,

$$\sum_{d|n} \sigma(d)\phi(n/d) = n\tau(n).$$

4. Let p be a prime which is $\equiv 3 \pmod 4$, and let $a \in \mathbb{Z}$. Show that

$$x^4 \equiv a \pmod{p}$$

is solvable if and only if $p|a$ or $(a/p) = 1$.

5. For an odd prime p, show that

$$\sum_{j=1}^{p-1} j^n \equiv \begin{cases} 0 \pmod{p} & \text{if } (p-1) \nmid n, \\ -1 \pmod{p} & \text{if } (p-1)|n. \end{cases}$$

6. Let $p = 4m+1$ be a prime, and let $d|m$. Show that

$$(d/p) = 1.$$